



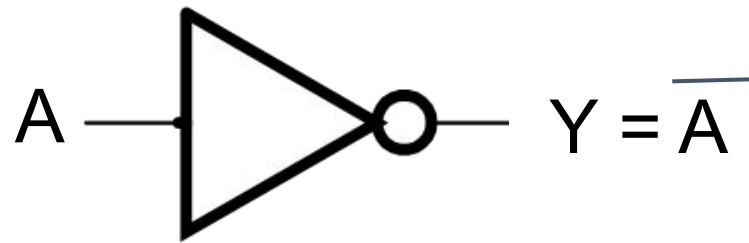
BITS Pilani
Pilani Campus

ITW Workshop Part-II (Teaching Demo)

CMOS Inverter

Ramesh Sambangi
EEE Department

Inverter

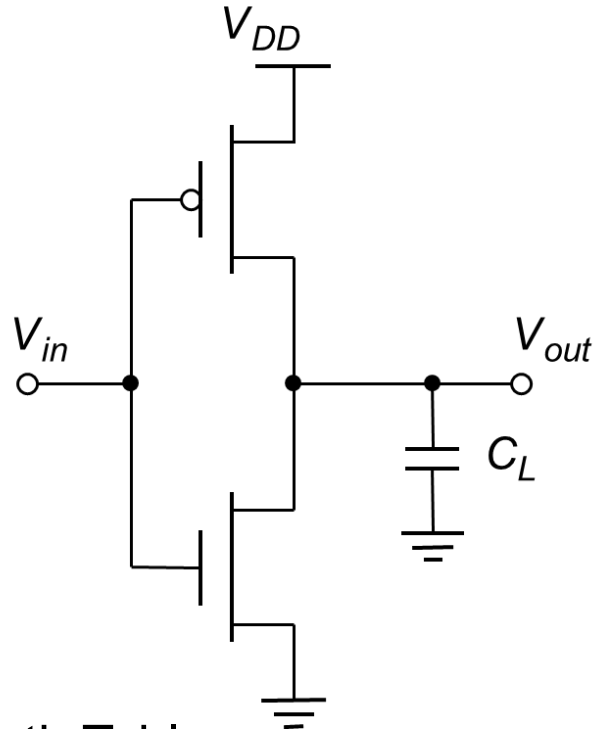
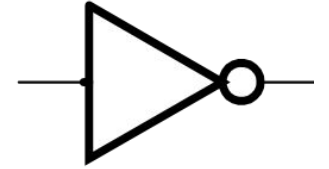


Truth Table

A	Y
0	1
1	0

An **inverter** is an electronic circuit that **flips** whatever signal it receives.

CMOS inverter



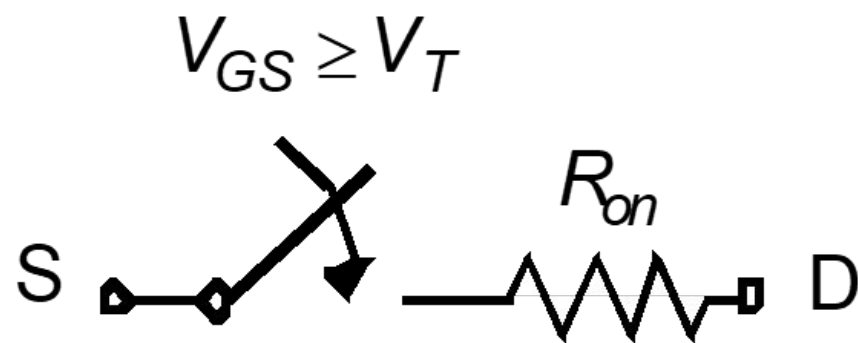
Truth Table

A	Y
0	1
1	0

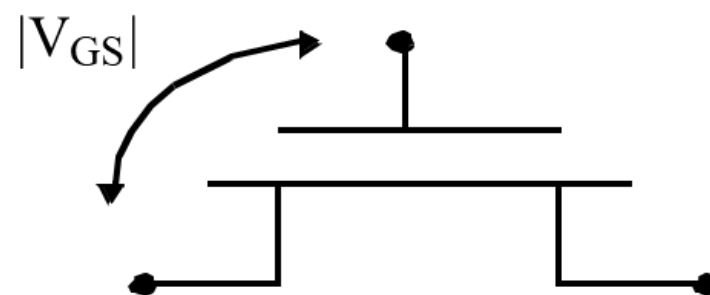
A CMOS inverter is the simplest and most important logic gate built using CMOS technology.

Feature	Explanation
Very low power	Almost no current flows except during switching
High noise immunity	Stable and reliable operation
High speed	Fast switching due to complementary action
Compact design	Needs only two transistors per inverter

A Switch!



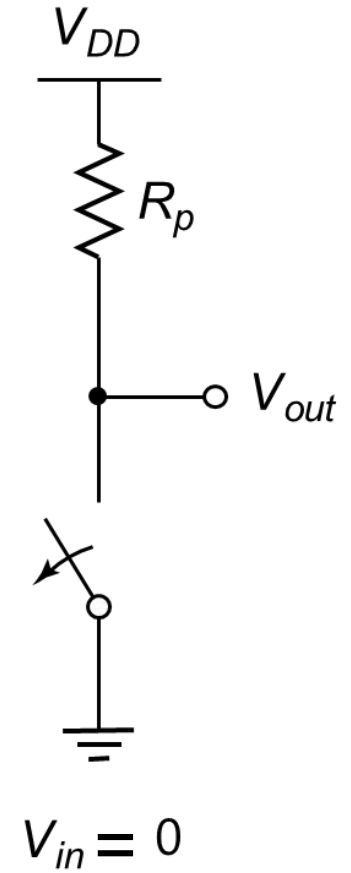
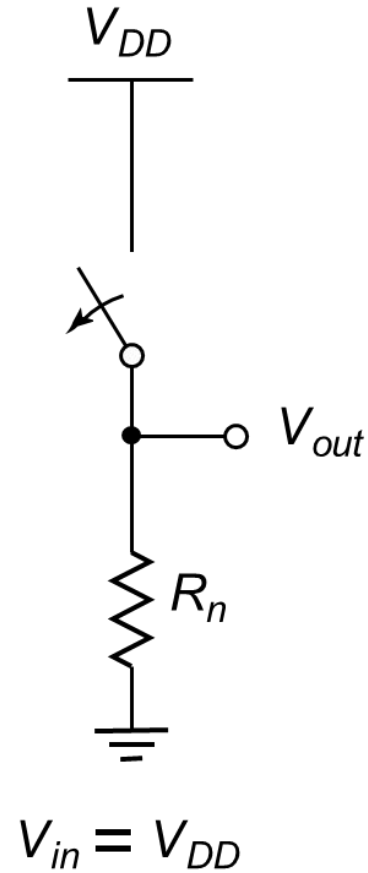
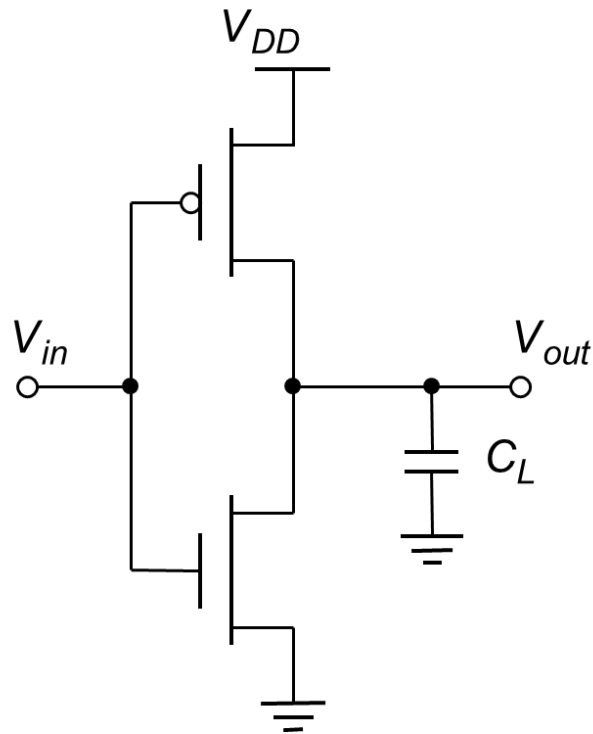
An MOS Transistor



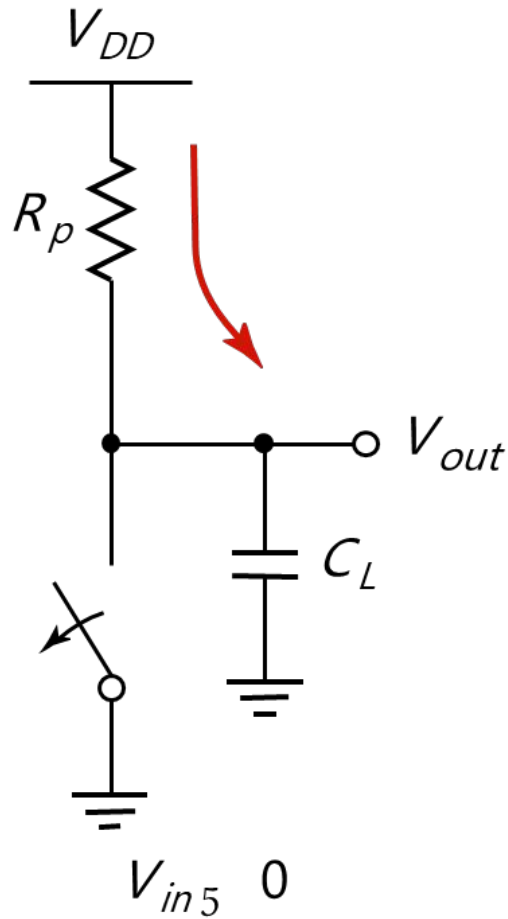
$$R_{eq} = \frac{1}{2} \left(\frac{V_{DD}}{I_{DSAT}(1 + \lambda V_{DD})} + \frac{V_{DD}/2}{I_{DSAT}(1 + \lambda V_{DD}/2)} \right) \approx \frac{3}{4} \frac{V_{DD}}{I_{DSAT}} \left(1 - \frac{5}{6} \lambda V_{DD} \right)$$

CMOS Inverter

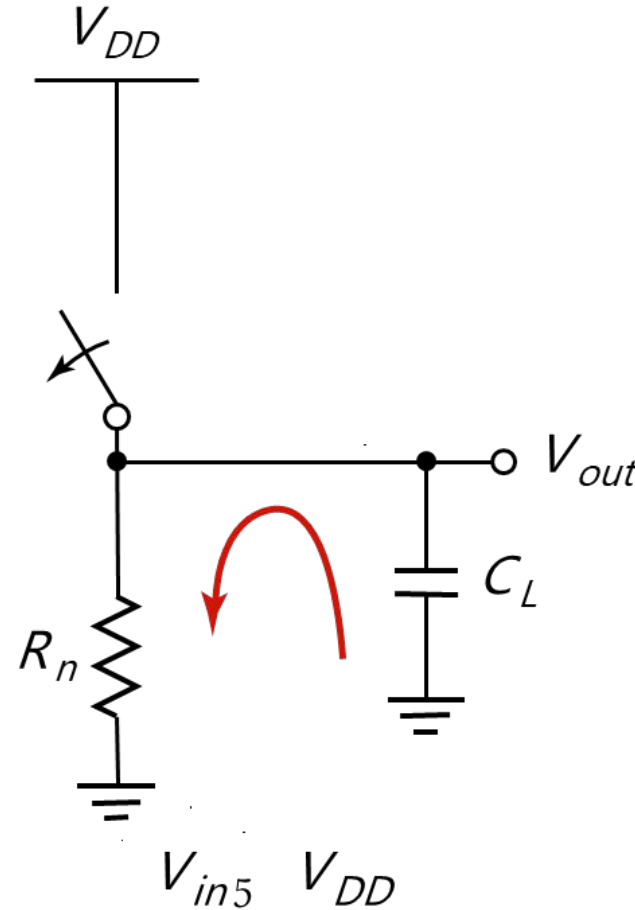
First-Order DC Analysis



CMOS Inverter: Transient Response



(a) Low-to-high

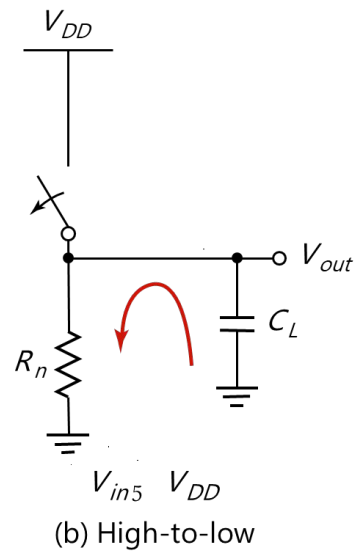
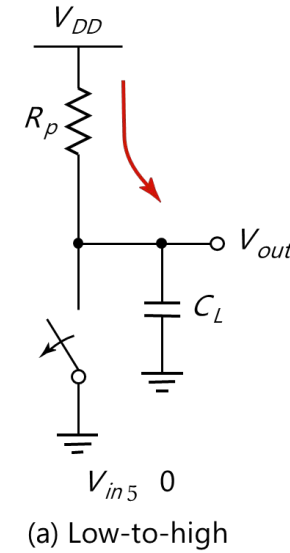
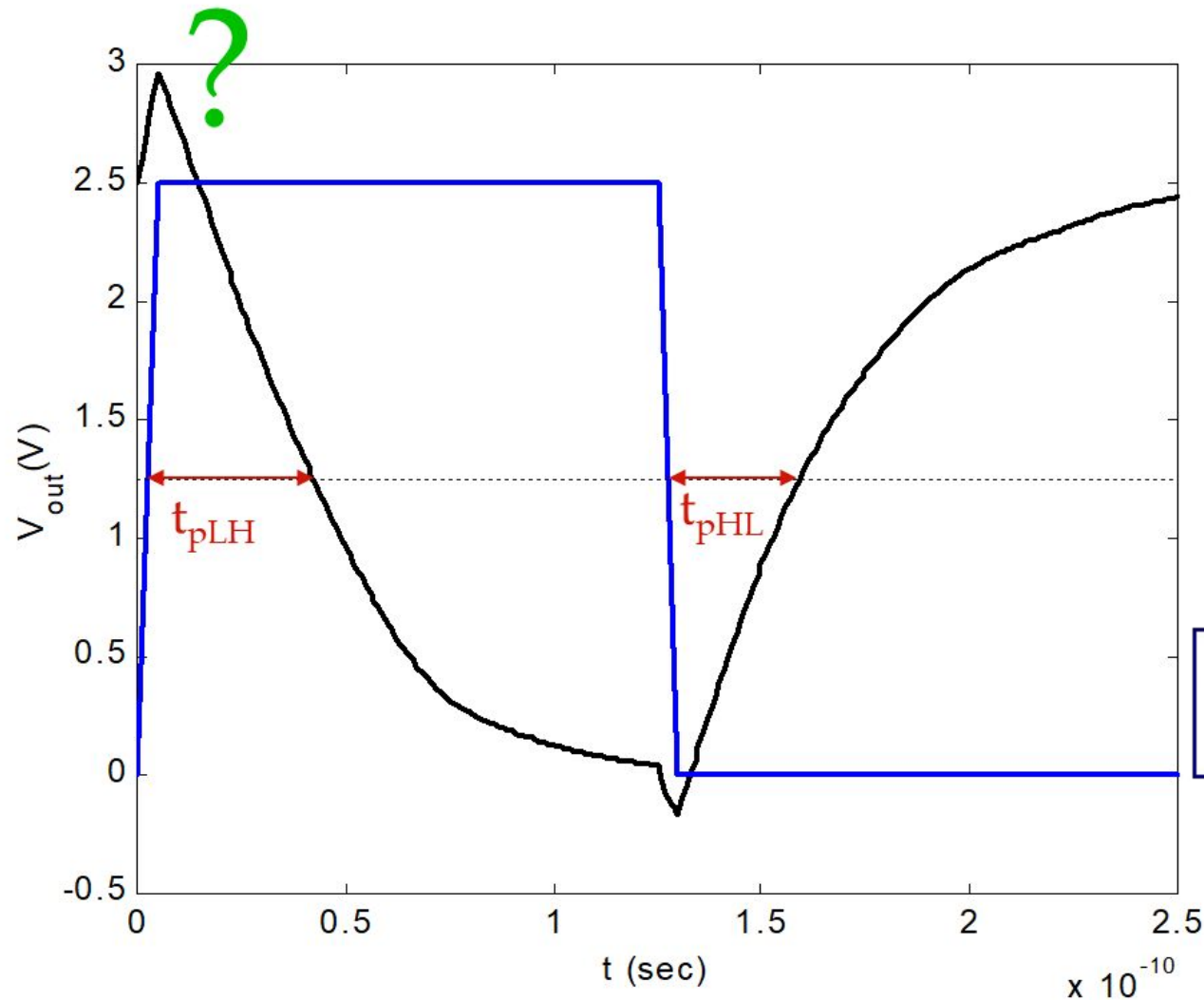


(b) High-to-low

$$t_{pLH} = 0.69 C_L R_{eqp}$$

$$t_{pHL} = 0.69 C_L R_{eqn}$$

Contd.



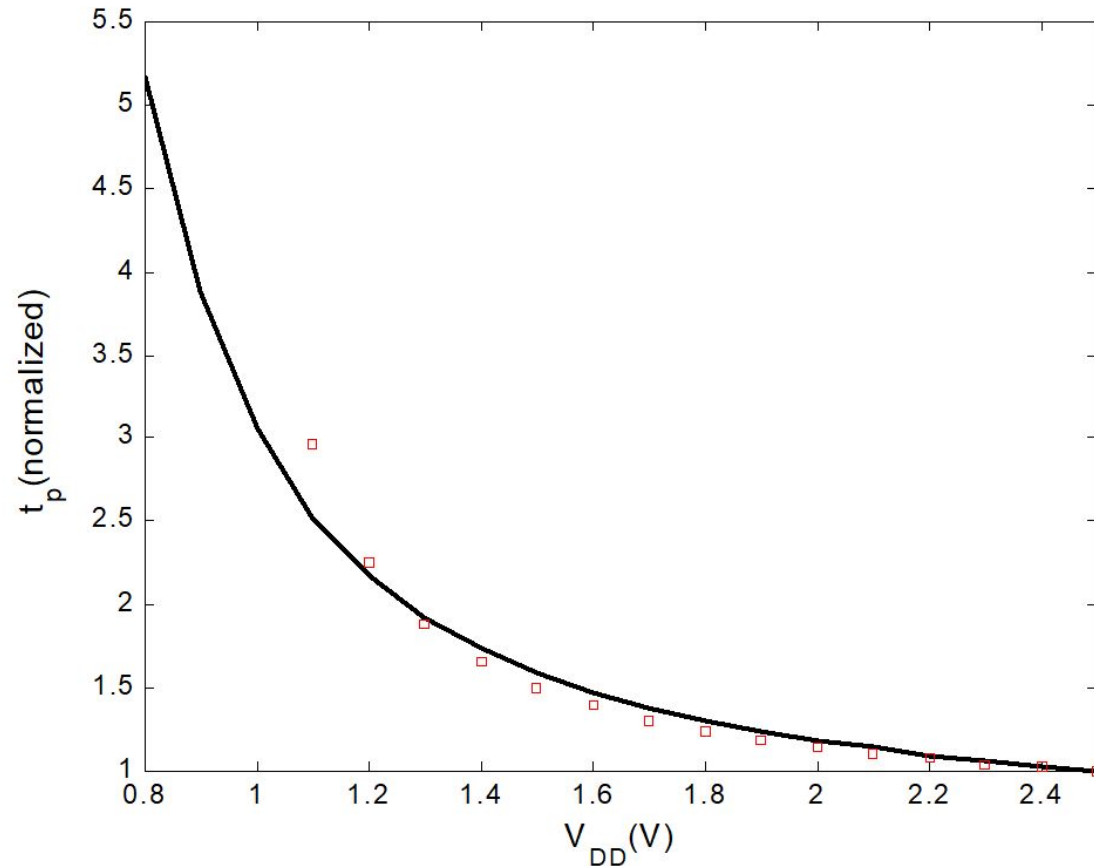
$$t_p = 0.69 C_L (R_{eqn} + R_{eqp})/2$$

Design for Performance



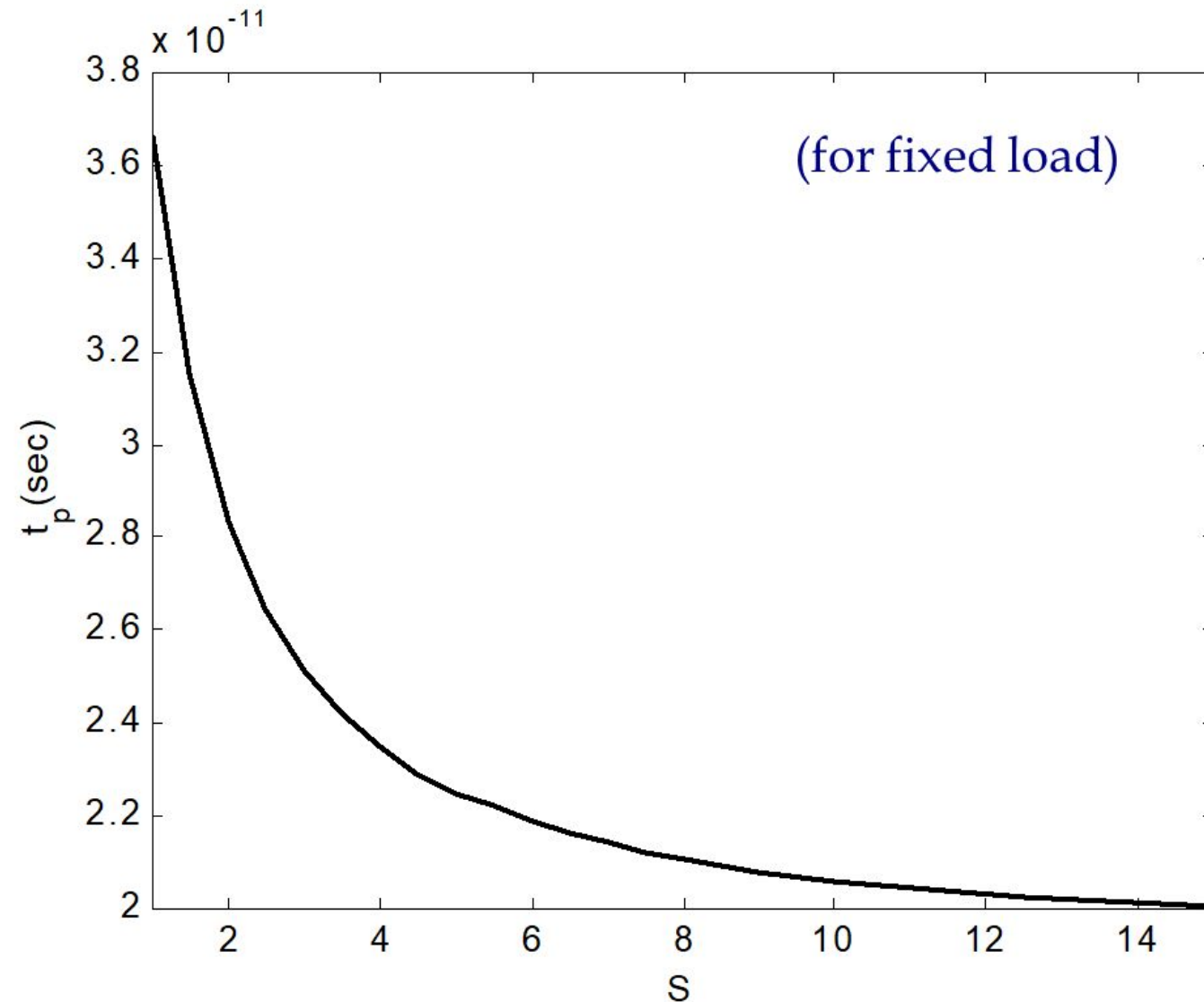
- ❑ Keep capacitances small
- ❑ Increase transistor sizes
 - watch out for self-loading!
- ❑ Increase V_{DD} (????)

Delay as a function of V_{DD}



$$R_{eq} = \frac{1}{2} \left(\frac{V_{DD}}{I_{DSAT}(1 + \lambda V_{DD})} + \frac{V_{DD}/2}{I_{DSAT}(1 + \lambda V_{DD}/2)} \right) \approx \frac{3}{4} \frac{V_{DD}}{I_{DSAT}} \left(1 - \frac{5}{6} \lambda V_{DD} \right)$$

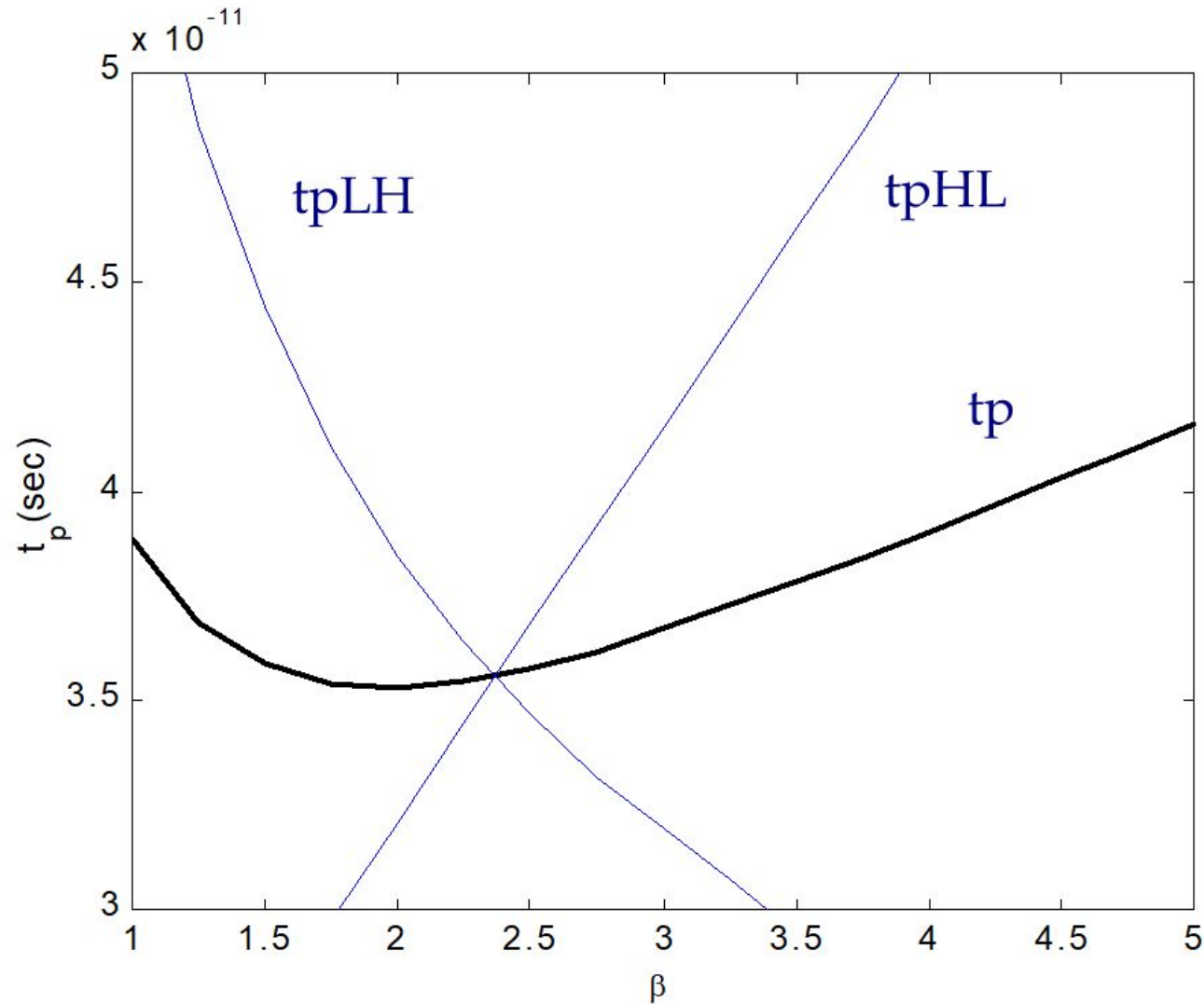
Device Sizing



$$t_p = 0.69 C_L (R_{eqn} + R_{eqp})/2$$

Self-loading effect:
Intrinsic capacitances
dominate

NMOS/PMOS ratio



$$\beta = W_p/W_n$$

$$t_p = 0.69 C_L (R_{eqn} + R_{eqp})/2$$

Power Dissipation



- **Dynamic Power Consumption**

Charging and Discharging Capacitors

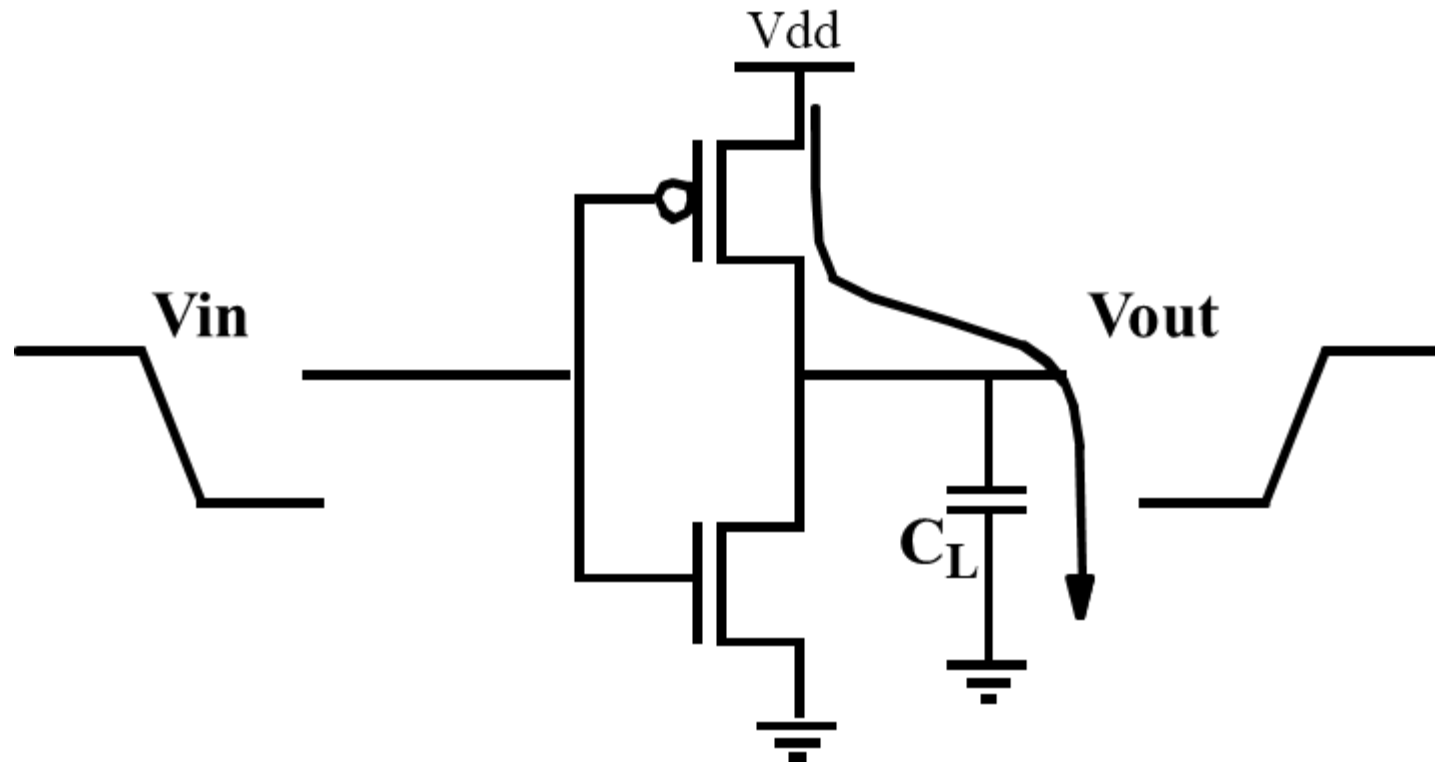
- **Short Circuit Currents**

Short Circuit Path between Supply Rails during Switching

- **Leakage**

Leaking diodes and transistors

Dynamic Power Dissipation



$$\text{Energy/transition} = C_L * V_{dd}^2$$

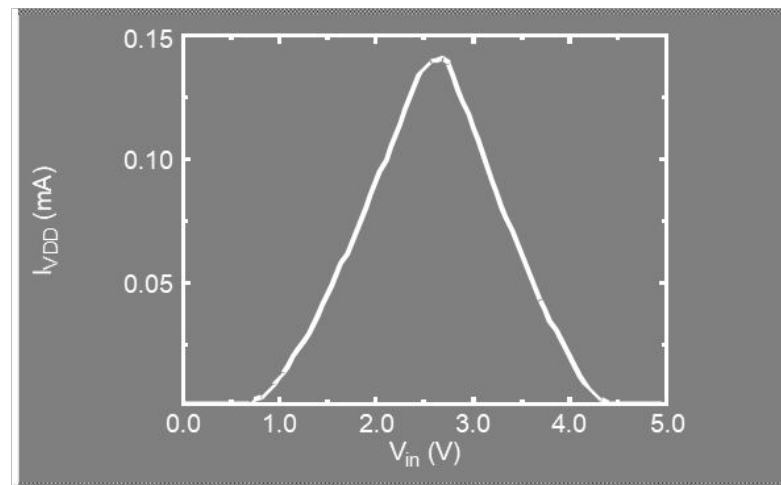
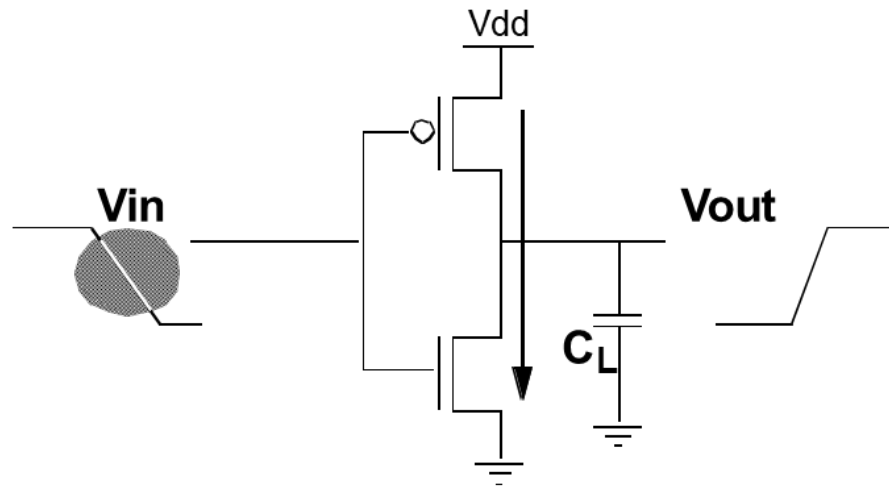
$$\text{Power} = \text{Energy/transition} * f = C_L * V_{dd}^2 * f$$

Short Circuit Currents

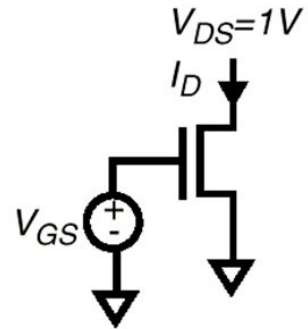
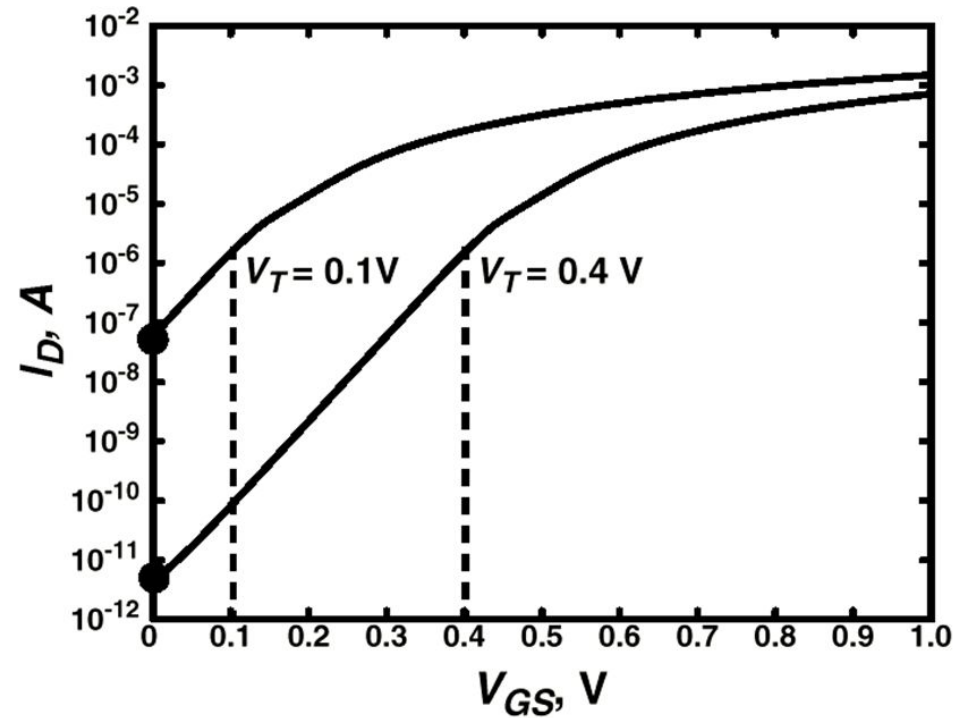
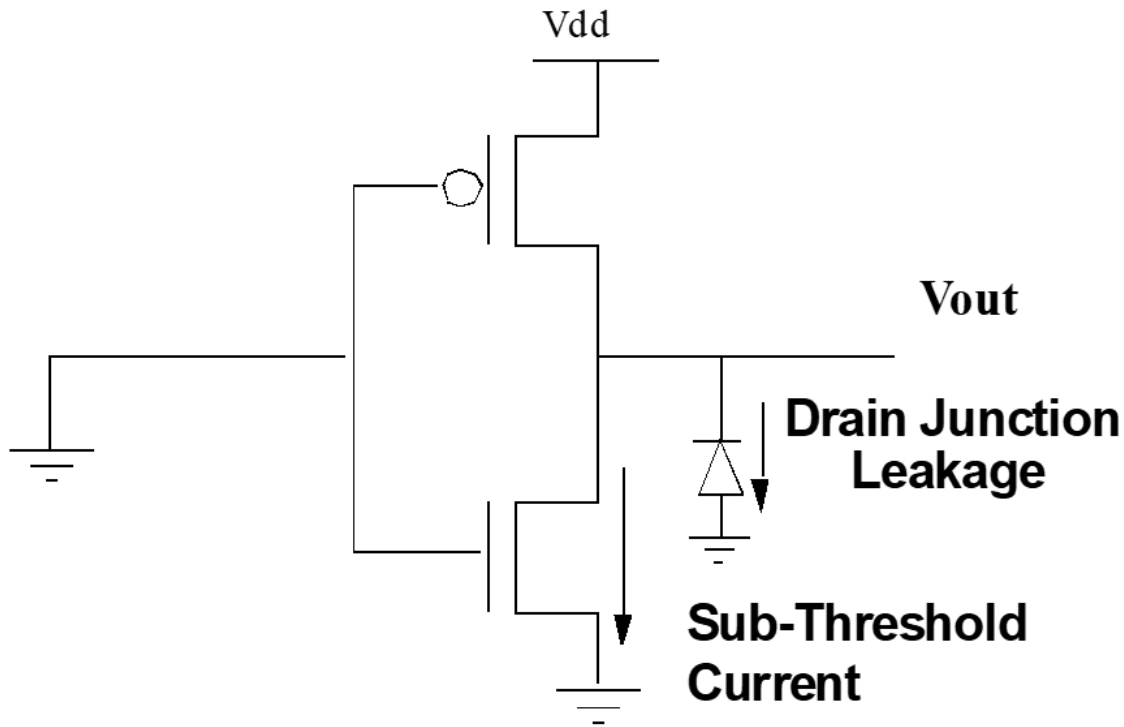
innovate

achieve

lead



Leakage



- Leakage control is critical for low-voltage operation

Principles for Power Reduction

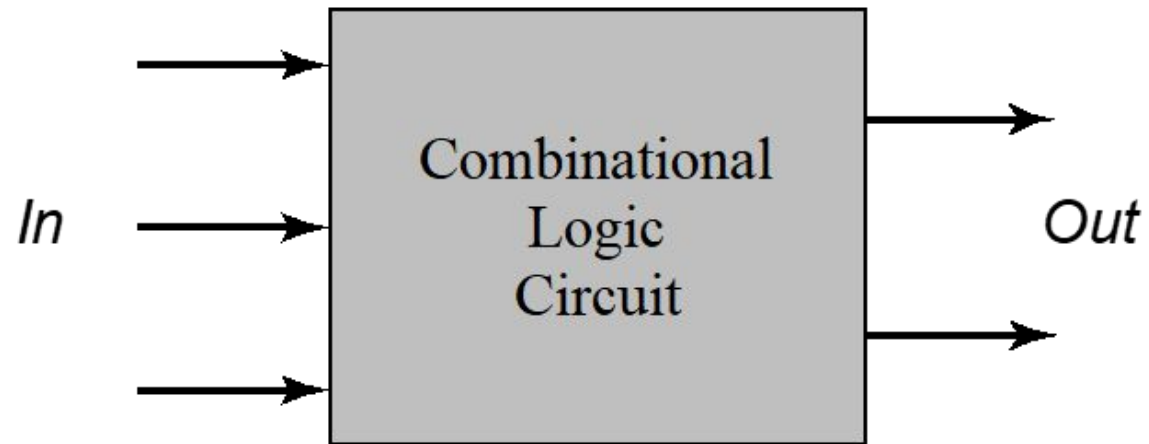


- ❑ Prime choice: Reduce voltage!
 - Recent years have seen an acceleration in supply voltage reduction
 - Design at very low voltages still open question (0.6 ... 0.9 V by 2010!)
- ❑ Reduce switching activity
- ❑ Reduce physical capacitance

$$\text{Energy/transition} = C_L * V_{dd}^2$$

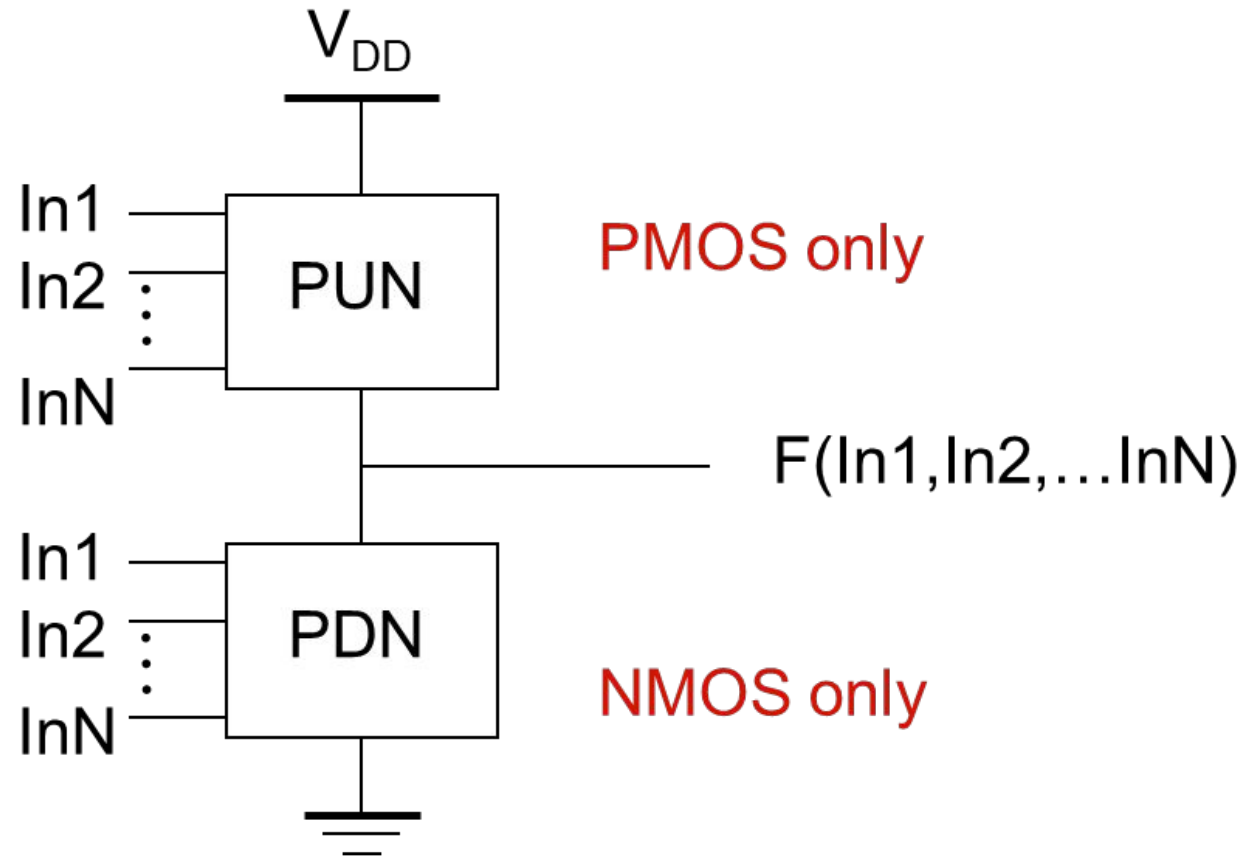
$$\text{Power} = \text{Energy/transition} * f = C_L * V_{dd}^2 * f$$

Combinational Logic Design



$$\text{Output} = f(\text{In})$$

Static Complementary CMOS



PUN and PDN are **dual** logic networks

Example Gate: NAND

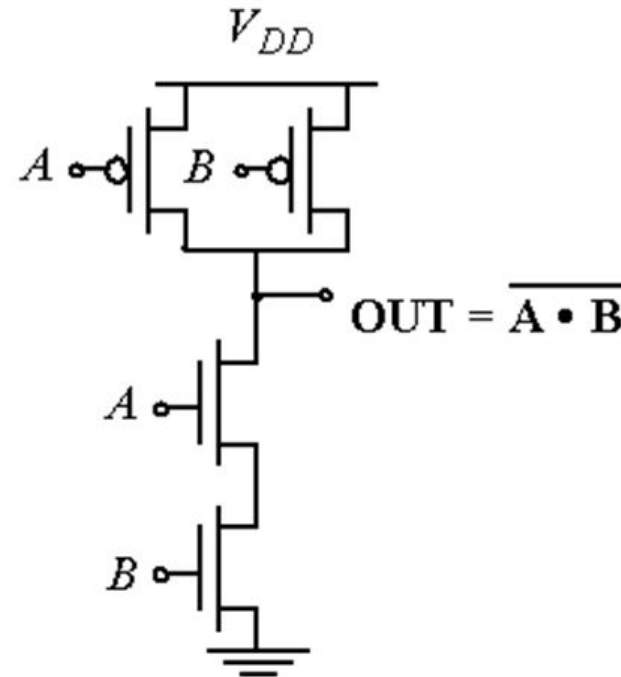
innovate

achieve

lead

A	B	Out
0	0	1
0	1	1
1	0	1
1	1	0

Truth Table of a 2 input NAND gate



PDN: $G = A B \Rightarrow$ Conduction to GND

PUN: $F = \overline{A} + \overline{B} = \overline{AB} \Rightarrow$ Conduction to V_{DD}

$$\overline{G(In_1, In_2, In_3, \dots)} \equiv F(\overline{In_1}, \overline{In_2}, \overline{In_3}, \dots)$$

Thank
You